



Assignment No. 10

Title : Writing data to internal storage in Android

Aim :

1. Introduction :

Internal storage in Android is used to store private data on the device memory. The files saved in internal storage are accessible only by the application and are deleted when the app is uninstalled.

Android provides different classes like `FileInputStream` and `FileOutputStream` to perform file handling operations such as reading and writing data.

2. File Handling in Android

File Handling allows an application to :

- store data permanently
- Read stored data when required
- maintain user-specific information securely

Types of storage in Android

1. Internal storage (Private)
2. External storage (Public)

3. Writing Data to Internal Storage

To write data into a file, Android provides the method :



openFileOutput()

- o used to create and write data into a file
- o file is stored in app's private directory.

Syntax:

```
FileOutputStream fos = openFileOutput ("filename.txt",  
MODE_PRIVATE);  
fos.write (data.getBytes());  
fos.close();
```

Explanation:

- o "filename.txt" → Name of file
- o MODE_PRIVATE → File is private to the app
- o write() → writes data
- o close() → closes the file

4) Reading data from internal storage

To read data, Android uses:

openFileInput()

Syntax:

```
FileInputStream fis = openFileInput ("filename.txt");  
int c;  
String temp = "";  
while ((c = fis.read()) != -1) {  
    temp = temp + Character.toString((char)c);  
}
```

3

```
fis.close();
```


Explanation :

- o read() → Reads data byte by byte
- o Loop continues until end of file (-1)
- o Data is converted into string format

5) File Input Stream and File Output Stream.

File Output Stream

- o used to write data into file
- o creates file if not exists
- o overwrites existing data (in mode PRIVATE)

File Input Stream

- o used to read data from file
- o reads data sequentially

6) Important File Handling Functions.

1. openFileOutput() : Opens file for writing in internal storage
2. openFileInput() : opens file for reading
3. write() : writes data into file
4. read() : Reads data from file
5. close() : closes file stream
6. deleteFile() : Deletes a file from internal storage
7. fileList() : Returns list of files in directory
8. getFilesDir() : Returns path of internal storage dir.
9. available() : Returns available bytes to read
10. flush() : clears output stream buffer.



7) Advantages of Internal Storage

- o secure (only app can access)
- o No permission required
- o data is always available
- o Automatically deleted on uninstall

8) Disadvantages

- o Limited storage space
- o Not accessible by other apps
- o Data lost after uninstall

9) conclusion

Internal storage is a secure and efficient way to store private data in Android applications. Using `FileInputStream` and `FileOutputStream`, developers can easily perform read and write operations.

It is suitable for storing sensitive data like user information, settings and app-related files.